Test Environment Summary

1. Machine Used for Testing

Component	Details
CPU	AMD EPYC 9754 (Zen 4c, family 25h), x86_64
CPU cores	32 vCPUs (1 socket, 32 cores, 1 thread/core); KVM virtualized
CPU features	AVX2, AVX-512 (incl. BF16, VNNI), AMD-V
GPU	NVIDIA RTX 6000 Ada Generation (AD102GL), Ada Lovelace architecture
GPU memory	48 GB (49,140 MiB); 300 W cap
Driver / CUDA	Driver 610.57.04, CUDA 13.3
System memory	62 GiB RAM (40 GiB available at capture); 4.9 GiB swap

Note: the GPU is shared; about 9.7 GiB was in use at capture time by vLLM and Triton Inference Server (38.8 GiB free).

2. Kubernetes Pods

Cluster: arcaai-dev-cluster | Namespace: project-hope-vox-infra | All pods Running, 1/1 replica, 0 restarts.

Pod	Image	Port	CPU (req / limit)	Memory (req / limit)	Storage (PVC)
postgres-pgvector-db	timescale/timescaledb-ha:pg18-all	5432	500m / 1	2Gi / 4Gi	64Gi (nfs-client)
silo-object-store	pgsty/silo (S3-compatible)	9000, 9001	500m / 500m	2Gi / 4Gi	150Gi (nfs-client)
vault-store	hashicorp/vault:1.18.3	8200	100m / 500m	128Mi / 512Mi	1Gi (default class)
redis-store	redis:8-alpine	6379	2000m / 2000m	4Gi / 4Gi	20Gi (nfs-client)

- postgres-pgvector-db: database "hope", StatefulSet, ClusterIP + NodePort 32251.
- silo-object-store: MinIO-compatible object store, ClusterIP + NodePorts 32631 (API) and 30372 (console).
- vault-store: Deployment (others are StatefulSets); audit flags single replica and missing readiness/liveness probes.
- redis-store: password-protected, append-only persistence enabled; NodePort 32522.